

Introduction

This model is a multimodal neural network designed to predict the future 3s trajectory of a vehicle. It processes visual data from two consecutive images with a temporal gap of 0.1s, along with vehicle speed derived from GPS data. The model outputs a sequence of 6 displacement vectors representing the future vehicle's path over 3 seconds.

1 Model Architecture

1. Image Processing Branch:

- **Convolutional Block 1:**
 - Conv2D: 64 filters, 3×3 kernel, padding=1, ReLU activation
 - BatchNorm2D
 - Conv2D: 64 filters, 3×3 kernel, padding=1, ReLU activation
 - BatchNorm2D
 - MaxPool2D: 2×2
- **Convolutional Block 2:**
 - Conv2D: 128 filters, 3×3 kernel, padding=1, ReLU activation
 - BatchNorm2D
 - Conv2D: 128 filters, 3×3 kernel, padding=1, ReLU activation
 - BatchNorm2D
 - MaxPool2D: 2×2
- **Convolutional Block 3:**
 - Conv2D: 256 filters, 3×3 kernel, padding=1, ReLU activation
 - BatchNorm2D
 - Conv2D: 256 filters, 3×3 kernel, padding=1, ReLU activation
 - BatchNorm2D
 - Conv2D: 256 filters, 3×3 kernel, padding=1, ReLU activation
 - BatchNorm2D
 - MaxPool2D: 2×2
- **Convolutional Block 4:**
 - Conv2D: 512 filters, 3×3 kernel, padding=1, ReLU activation
 - BatchNorm2D
 - Conv2D: 512 filters, 3×3 kernel, padding=1, ReLU activation
 - BatchNorm2D
 - Conv2D: 512 filters, 3×3 kernel, padding=1, ReLU activation
 - BatchNorm2D
- **Spatial Attention:**
 - Conv2D: 1 filter, 1×1 kernel, Sigmoid activation
 - Attention application through element-wise multiplication
- **Feature Concatenation:**
 - Concatenate features from both images (creating a 1024-channel feature map)
- **Final Convolutional Processing:**
 - Conv2D: 1024 filters, 3×3 kernel, padding=1, ReLU activation
 - Conv2D: 1024 filters, 1×1 kernel, ReLU activation
 - Global Average Pooling to create a 1024-dimensional vector

2. Dynamic Input Processing Branch (processes vehicle speed):

- Dense layer with 32 units, Tanh activation, input dimension 1

- Batch normalization

3. Feature Fusion and Fully Connected Layers:

- Concatenation of image features (1024) and dynamic features (32)
- Dense layer with 2048 units, ReLU activation
- Dropout (0.5)
- Dense layer with 1024 units, ReLU activation
- Dropout (0.5)
- Dense layer with 512 units, ReLU activation
- Dropout (0.3)
- Dense layer with 128 units, ReLU activation
- Dense layer with 12 units (output)
- Reshape to (6, 2) format - representing 6 vectors with x,y coordinates